PENETRATION TESTING REPORT

Meridian SE API Assessment – Layer 1

1. Candidate Information

Candidate Name: Jagath Kalyan Boddeti
Assessment Type: API Security Assessment
Environment Used: Kali Linux, Python 3, Postman
Assessment Status: Layer 1 Successfully Solved

2. Introduction

This report explains the methodology and process followed to complete Layer 1 of the Meridian SE API Assessment. The assessment focused on API analysis, dataset collection, integrity verification, automation scripting, and problem solving. The report is written in simple English so both technical and non-technical readers can understand the process clearly.

3. Tools Used

Tool	Purpose
Kali Linux	Testing environment
Python 3	Automation scripting
Requests Library	API requests
Hashlib	SHA256 integrity hashing
Postman	API testing and authentication
VS Code	Script development

4. Methodology

Phase 1 – API Reconnaissance

The API endpoints were explored using Postman and Python scripts to understand how the service worked.

Phase 2 – Authentication Testing

Bearer Token Authentication was used for accessing authenticated endpoints. The API key was added in Postman and Python request headers.

Phase 3 – Dataset Collection

Pagination and automation scripts were used to collect all encrypted records from the API.

Phase 4 – Integrity Hash Analysis

Several SHA256 hashing approaches were tested until the correct validation format was identified.

Phase 5 – Layer 1 Validation

The final SHA256 integrity hash was submitted successfully and Layer 1 was solved.

5. Assessment Evidence

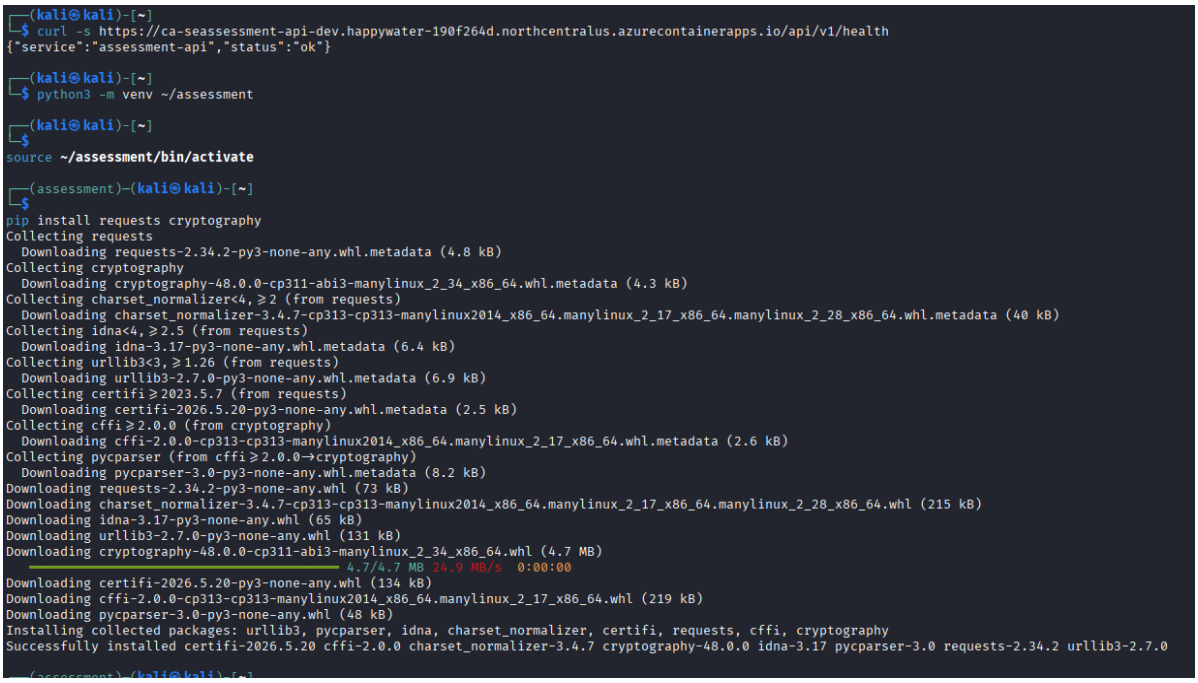

Figure 1: Environment setup and Python virtual environment configuration.

Figure 2: API endpoint enumeration and reconnaissance testing.

Figure 3: Dataset extraction and batch retrieval process.

Figure 4: SHA256 integrity hash testing and analysis.

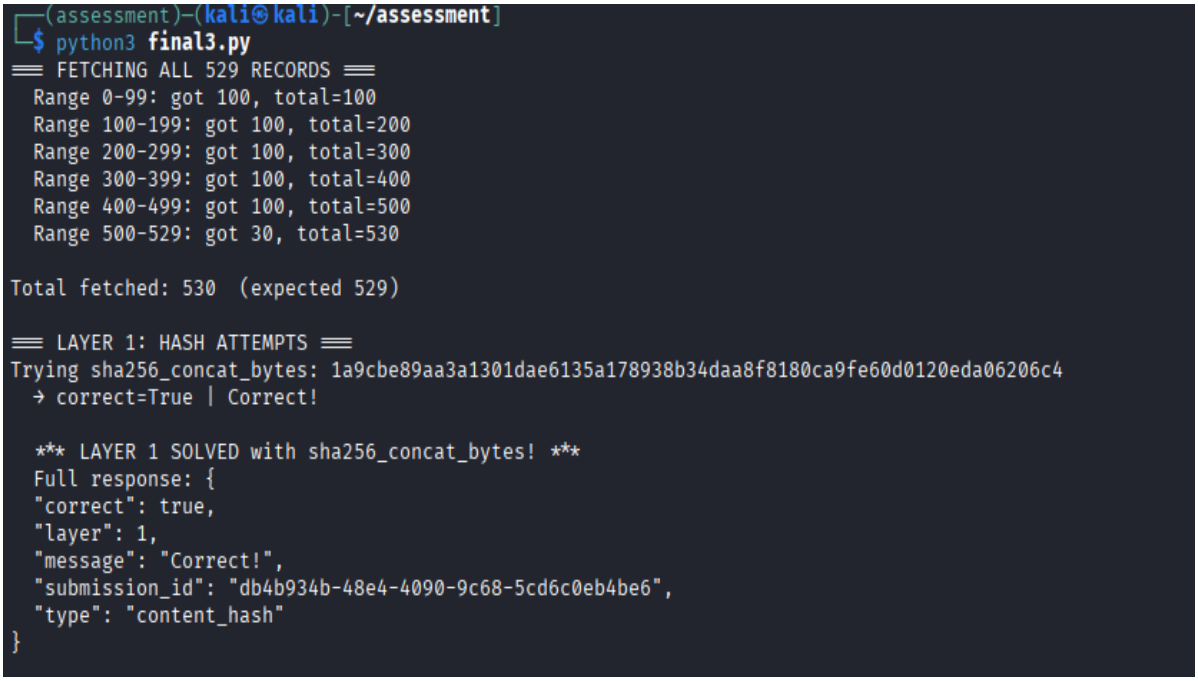

Figure 5: Successful Layer 1 validation response.

6. Scripts Executed During Assessment

The following Python scripts were developed and executed during the assessment: - explore.py → endpoint enumeration and reconnaissance - solve.py → dataset extraction and pagination handling - solve2.py → integrity hash analysis - final1.py → advanced hashing attempts - final2.py → cryptographic testing - final3.py → final Layer 1 validation workflow These scripts automated the assessment process and improved testing efficiency.

7. Challenges Faced

Several challenges were encountered: - incorrect dataset counts, - API rate limiting, - failed hash attempts, - authentication header issues, - debugging multiple hashing methods. These problems were solved through repeated testing and script improvements.

8. Conclusion

Layer 1 of the Meridian SE API Assessment was successfully completed through structured analysis, automation scripting, API testing, and integrity verification. The assessment demonstrated practical cybersecurity skills including API analysis, debugging, cryptographic validation, and problem solving.